

Experiment 02: Input & Output Manual

Password Strength Indicator and Star Rating System using jQuery

🎓 Student & Laboratory Credentials

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- **Academic Year:** 2026 – 2027
- **Course:** Skill Based Lab - Advanced Web Technology (SBL-AWT)
- **Institution:** Ramrao Adik Institute of Technology (RAIT), D Y Patil Deemed to be University, Nerul, Navi Mumbai
- **Problem Statement Domain:** HealthPulse — Digital Healthcare & Clinical Management Portal
- **Experiment Title:** Write a Program to Create and Build a Password Strength Indicator and Star Rating System using jQuery

1. Experiment Aim

To design and build an interactive, client-side web application implementing: 1. A real-time **Password Entropy & Strength Analyzer** with visual checklist feedback for healthcare staff credentials. 2. A dynamic **5-Star Doctor Consultation Rating System** with hover preview, score locking, and animated submission confirmation using the jQuery library.

2. 🖱️ Input Specification

2.1 File System Input

File Name	Format	Role & Implementation Responsibility
index.html	HTML5	Two-column layout containing Password Strength card (input, checklist, meter) and Doctor Rating card (stars, feedback textarea).
style.css	CSS3	Interactive progress bar styling, star transitions, amber glow animations, and responsive flex container rules.
script.js	jQuery (v3.7.1)	Event listeners on <code>input</code> , <code>keyup</code> , <code>mouseenter</code> , <code>mouseleave</code> , <code>click</code> , regex scoring, DOM class swapping, and <code>.fadeIn()</code> animations.

2.2 Execution Command Input

```
# Navigate to the experiment directory
cd /home/ubuntu/SBL-EXPT/Asmit_25CE1051/Expt-02-Password-Rating-jQuery

# Launch in browser directly
chromium-browser index.html

# Or run via local server
python3 -m http.server 8081
```

2.3 Password Evaluation Rules & Scoring Input Matrix

The jQuery analyzer evaluates password input across 5 independent criteria:

RULE #	CRITERIA DESCRIPTION	REGEX PATTERN / JQUERY CONDITION	WEIGHT / SCORE
Rule 1	Minimum 8 Characters Length	val.length >= 8	+1 Point (20%)
Rule 2	Lowercase Alphabet (a-z)	/[a-z]/.test(val)	+1 Point (20%)
Rule 3	Uppercase Alphabet (A-Z)	/[A-Z]/.test(val)	+1 Point (20%)
Rule 4	Numeric Digit (0-9)	/[0-9]/.test(val)	+1 Point (20%)
Rule 5	Special Symbol	/[!@#\$%^&*()_+~\-=\[\]\{\};':"\ ,.<>\?]/	+1 Point (20%)

2.4 Doctor Consultation Rating Input Scale

STAR LEVEL	RATING SCORE	DISPLAY LABEL TEXT	COLOR CODE	CLINICAL FEEDBACK CONTEXT
☆☆☆☆	1 Star	Poor Experience	#ef4444 (Red)	Long wait time, inadequate care
☆☆☆☆	2 Stars	Fair / Average	#f97316 (Orange)	Basic treatment, delayed tests
☆☆☆☆	3 Stars	Good	#eab308 (Yellow)	Satisfactory consultation
☆☆☆☆	4 Stars	Very Good	#84cc16 (Lime)	Attentive doctor, clear advice
☆☆☆☆	5 Stars	Exceptional Experience	#10b981 (Emerald)	Exemplary clinical excellence

2.5 Sample Test Input Vectors

Password Strength Test Data:

- Input 1: `adm` (Length < 8, lowercase only)
- Input 2: `healthpulse` (Length >= 8, lowercase only)
- Input 3: `HealthPulse` (Length >= 8, lowercase + uppercase)
- Input 4: `HealthPulse2026` (Length >= 8, lowercase + uppercase + number)
- Input 5: `HealthPulse@2026!` (Length >= 8, lowercase + uppercase + number + symbol)

Star Rating Test Data:

- Action: Hover on 4th Star → Click on 5th Star
- Reviewer Name: `Asmit Jogdand (Patient ID: HP-2026-9921)`
- Doctor Selected: `Dr. Amarsinh V. Vidhate (Chief Medical Director - Cardiology)`
- Written Review: `"Dr. Vidhate provided prompt and thorough cardiac diagnostics. The clinical staff demonstrated highest standards of professionalism."`

3. Output Specification

3.1 Graphical User Interface Output Representation

```
+-----+
| [HealthPulse Portal] SBL - Advanced Web Technology (Sem VI) Student: Asmit (25CE1051) |
+-----+
|
| +-- MODULE 1: Credential Security Analyzer --+ +-- MODULE 2: Doctor Consultation Rating -----+ |
| | Enter Staff Password: | | Physician: Dr. Amarsinh V. Vidhate (Cardiology)| | | |
| | [ HealthPulse@2026! ] [👁️] | | Overall Patient Satisfaction Rating: | |
| | | | | | | |
| | Password Strength Meter: | | [ * ] [ * ] [ * ] [ * ] [ * ] | |
| | [=====] 100% | | (Glowing Amber Gold *****) | |
| | Status: Very Strong (HIPAA Compliant) | | Selected Rating: 5 / 5 – Exceptional | |
| | | | | |
| | Security Rule Checklist: | | Patient Comments & Suggestions: | |
| | [✓] Minimum 8 characters in length | | [ Dr. Vidhate provided prompt and thorough ] | |
| | [✓] At least one lowercase letter (a-z) | | [ cardiac diagnostics. Highly recommended. ] | |
| | [✓] At least one uppercase letter (A-Z) | | | |
| | [✓] At least one numeric digit (0-9) | | [ Submit Review ] [ 🔄 Reset ] | |
| | [✓] At least one special symbol (@#$!%) | | | |
| +-----+ +-----+
+-----+
```

3.2 Submission Confirmation Banner Output (jQuery Animated)

Upon clicking **"Submit Review"**, jQuery fires `.fadeIn(400)` to render the persistent review receipt:

```
+-----+
|  ▫ DOCTOR CONSULTATION REVIEW SUBMITTED SUCCESSFULLY |
+-----+
| Review ID:          REV-HP-2026-88192 |
| Consulting Specialist: Dr. Amarsinh V. Vidhate (Chief Medical Director) |
| Awarded Rating:     ***** 5.0 / 5.0 (Exceptional Experience) |
| Patient Feedback:    "Dr. Vidhate provided prompt and thorough cardiac |
|                      diagnostics. Highly recommended." |
| Verified Timestamp:  2026-10-01 10:20:45 IST |
| Status:             Published to Clinical Quality Board |
+-----+
```

3.3 Test Case Execution Results

Test Scenario	Input Given	jQuery DOM Execution & Calculation	Output Result Observed	Status
Empty Input	<code>""</code>	All regex tests evaluate <code>false</code>	Score 0%, Gray Bar (<code>#64748b</code>), All 5 rules show <code>[x]</code>	PASS
Short Lowercase	<code>adm</code>	Length < 8, Lowercase <code>true</code>	Score 20%, Red Bar (<code>#ef4444</code>), Label: "Very Weak"	PASS
Length Only	<code>healthpulse</code>	Length <code>true</code> , Lowercase <code>true</code>	Score 40%, Orange Bar (<code>#f97316</code>), Label: "Weak"	PASS
Alpha Mixed	<code>HealthPulse</code>	Length, Lower, Upper <code>true</code>	Score 60%, Yellow Bar (<code>#eab308</code>), Label: "Medium"	PASS
Alphanumeric	<code>HealthPulse2026</code>	4 conditions <code>true</code>	Score 80%, Lime Bar (<code>#84cc16</code>), Label: "Strong"	PASS
Full Entropy	<code>HealthPulse@2026!</code>	All 5 conditions <code>true</code>	Score 100%, Emerald Bar (<code>#10b981</code>), Label: "Very Strong"	PASS

Test Scenario	Input Given	jQuery DOM Execution & Calculation	Output Result Observed	Status
Password Unmask	Click eye icon 	<code>type</code> toggled <code>password</code> \$ \\leftrightarrow\$ <code>text</code>	Obfuscated dots reveal plain text characters	PASS
Star Hover	Hover 4th star	<code>mouseenter</code> fires, adds <code>.hovered</code> to 1..4	4 stars glow amber, label reads "Very Good (4 Stars)"	PASS
Star Click Lock	Click 5th star	<code>click</code> locks <code>selectedRating = 5</code>	All 5 stars locked <code>.active</code> , persistent amber fill	PASS
Empty Review Submit	No stars selected	Validation guard blocks empty submission	Browser warning alert: "Please select a star rating first!"	PASS

4. Conclusion & Verification

Experiment 02 was successfully verified. The jQuery event pipeline (`input keyup`, `mouseenter`, `mouseleave`, `click`) delivered real-time, responsive feedback. The password entropy calculator accurately enforced hospital staff credential security, and the star rating widget enabled smooth review submissions.